Weapon-Target Assignment Problem

CSE 462: Algorithm Engineering Sessional

Checkpoint 1 Presentation

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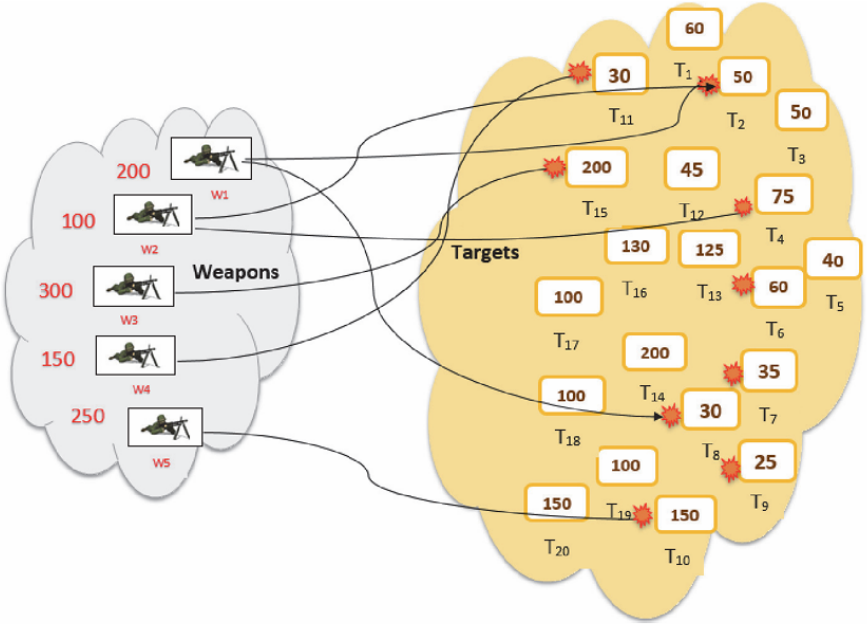
Real-World Motivation: WTA in Action

Air Defense (Iron Dome)



Ground Attack Scenario

WTA Mapping:

- Interceptor Missiles = **Weapons**
- Incoming Rockets = **Targets**
- → Assignment = **WTA Solution**

WTA Mapping:

- W1–W5 (with costs) = **Weapons**
- T1–T20 (with values) = **Targets**
- → Arrows = **Assignments**

Presentation Outline

Part 1: Theory

1. **Problem Definition**
   - Formal definition
   - Mathematical formulation
   - Classifications
2. **NP-Hardness Proofs**
   - Reduction from SUBSET PRODUCT
   - Reduction to Multiple Knapsack
3. **Algorithm Survey**
   - Exact, approximation, heuristic algorithms

Part 2: Implementation

4. **Selected Algorithms**
   - Maximum Marginal Return (MMR)
   - Ant Colony Optimization (ACO)
5. **Experimental Design**
   - Benchmark instances
   - Metrics and analysis plan
6. **Applications**
   - Military and civilian uses

What is the Weapon-Target Assignment Problem?

A fundamental combinatorial optimization problem in operations research

Problem Definition

Weapon-Target Assignment (WTA) Problem

The WTA problem involves the **strategic allocation of a set of weapons to a set of targets** to achieve a specific defensive or offensive objective.

Core Challenge: Stochastic Nature of Combat

Every engagement is modeled as a **probabilistic event** where a weapon has a specific likelihood of destroying its assigned target.

Computational Complexity

- The problem is mathematically proven to be **NP-complete**
- As the number of weapons and targets grows, complexity increases **exponentially**
- Finding an absolute optimal solution becomes extremely difficult for large instances

Classification: Asset-based vs. Target-based

Two Perspectives of WTA

The WTA problem can be viewed from two distinct perspectives depending on whether the objective prioritizes **destruction of enemy forces** or **protection of friendly resources**.

Target-based Perspective

Values assigned to each **enemy target** based on threat they pose.

- **Goal:** Maximize total damage inflicted on incoming targets
- **Math: Minimize** expected survival value of targets
- (Equivalent to maximizing damage)

Asset-based Perspective

Values assigned to **defended assets** (cities, bases) rather than targets.

- **Goal:** Maximize combined survival value of friendly assets
- **Requirement:** Map which targets attack which assets
- **Math: Maximize** expected value of surviving assets

Mathematical Representation: Parameters and Variables

Target-based Formulation – Parameters

- $W = \{1, \ldots, m\}$: The set of available **weapon types**
- $T = \{1, \ldots, n\}$: The set of **targets** to be engaged
- v_j: The **strategic military value** or importance of target j
- p_{ij}: The "**probability of kill**" – chance that one weapon of type i will destroy target j
- w_i: The **total quantity** of weapons of type i available

Decision Variable

$$x_{ij} \in \{0, 1, 2, \ldots\}$$

Integer variable representing the number of weapons of type i assigned to target j

Mathematical Representation: Objective Functions

Goal: Maximize damage (or equivalently, minimize survival)

While the goal is to maximize damage, the standard convention is to minimize remaining survival value.

1. Maximizing Total Expected Damage

$$\text{Max } Z = \sum_{j=1}^{n} v_j \left[1 - \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}} \right]$$

2. Minimizing Expected Survival Value

$$\text{Min } Z = \sum_{j=1}^{n} v_j \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$$

Key Insight

$\prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$ = **Probability that target j survives ALL assigned weapons**
The product captures the **combined effect** of multiple weapons attacking the same target.

Mathematical Representation: Constraints

Constraint 1: Resource Availability

$$\sum_{j=1}^{n} x_{ij} \leq w_i, \quad \forall i \in W$$

Meaning: Total weapons of type i assigned across all targets cannot exceed available quantity w_i

Constraint 2: Integrality

$$x_{ij} \in \{0, 1, 2, \dots\}$$

Meaning: We can only assign **whole numbers** of weapons (discrete assignments)

Complete Formulation Summary

Minimize $\sum_j v_j \prod_i (1 - p_{ij})^{x_{ij}}$ **s.t.** $\sum_j x_{ij} \leq w_i, \quad x_{ij} \in \mathbb{Z}^+$

Linear vs. Non-Linear Assignment

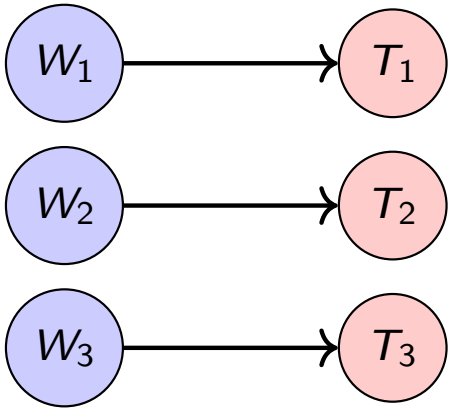
Linear Assignment Problem (LAP)

- **Structure:** One weapon assigned to exactly one target (**One-to-One**)
- **Linearity:** Objective function is linear
- Assignments are **independent**
- **Polynomial time solvable** (Hungarian algorithm)

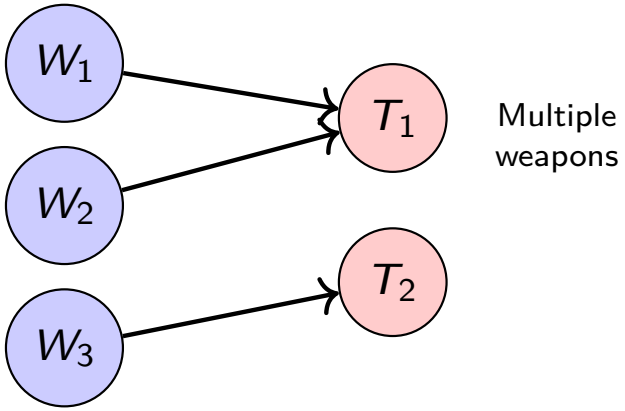
Non-Linear Assignment (Non-LAP)

- **Structure:** Multiple weapons can be assigned to single target (**Many-to-One**)
- **Non-Linearity:** Due to "**diminishing returns**"
- Additional weapons = **overkill** effect
- **NP-complete!**

LAP (One-to-One)



Non-LAP (Many-to-One)



Polynomial-Time Reductions

Proving WTA is NP-complete through formal reductions

WTA Decision Problem

Decision Version of WTA

Given:

- Survival probability $q_{i,j} = 1 - p_{i,j}$
- Threshold value k

Question: Does there exist an assignment $x_{i,j} \in \{0, 1\}$ such that:

$$D = \sum_{j=1}^{n} V_j \prod_{i=1}^{m} q_{i,j}^{x_{i,j}} \leq k$$

Subject to Constraints

$$\sum_{j=1}^{n} x_{i,j} \leq W_i \quad \text{for } i = 1, \ldots, m$$

$$x_{i,j} \in \{0, 1\}$$

where $x_{i,j} = 1$ if weapon i assigned to target j, and $x_{i,j} = 0$ otherwise.
D represents **total expected damage** (weighted survival probability of all targets)

Two Polynomial-Time Reductions

Our Approach

We demonstrate two polynomial-time reductions:

Reduction 1

SUBSET PRODUCT $\leq_p$ WTA

$\Rightarrow$ Proves **WTA is NP-hard**

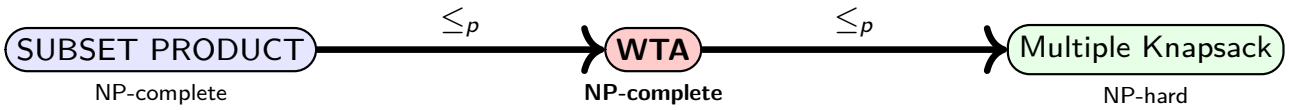
(Reduce FROM a known NP-complete problem TO WTA)

Reduction 2

WTA $\leq_p$ Multiple Knapsack

$\Rightarrow$ Shows WTA reduces to another known NP-hard problem

(Shows connection between problems)



Reduction 1: SUBSET PRODUCT $\leq_p$ WTA

SUBSET PRODUCT Problem (NP-complete)

Given: Set $A = \{a_1, a_2, \ldots, a_m\}$ and target product b
Find: Subset $X \subseteq A$ where $\prod_{x \in X} a_x = b$

WTA Instance Construction

Let $P = \prod_{i=1}^{m} a_i$. Construct WTA with:

- **Targets:** 2 targets with values $V_1 = V_2 = P$
- **Weapons:** $m + 2$ weapons: $W_1, W_2, \ldots, W_m, W_\alpha, W_\beta$
- **Survival probabilities:**
  - $q_{i,1} = q_{i,2} = \frac{1}{a_i}$ for $i = 1, \ldots, m$
  - $q_{\alpha,1} = q_{\alpha,2} = \frac{b}{2P}$
  - $q_{\beta,1} = q_{\beta,2} = \frac{1}{2b}$
- **Threshold:** $k = 1$ (we ask if $D \leq 1$)

Time Complexity

Constructs WTA with 2 targets and $m + 2$ weapons in $O(m)$ time – **Polynomial!**

Reduction 1: Forward Direction

Forward ($\Rightarrow$): SUBSET PRODUCT has solution $\Rightarrow$ WTA has solution

Suppose $X \subseteq \{1, \ldots, m\}$ with $\prod_{x \in X} a_x = b$. Let $Y = \{1, \ldots, m\} \setminus X$.
Then:

$$\prod_{i \in X} \frac{1}{a_i} = \frac{1}{b}, \quad \prod_{y \in Y} \frac{1}{a_y} = \frac{b}{P}$$

Assignment: $\{W_x : x \in X\} \cup \{W_\alpha\} \to T_1$ and $\{W_y : y \in Y\} \cup \{W_\beta\} \to T_2$
Total damage:

$$\begin{aligned} D &= V_1 \cdot \frac{b}{2P} \cdot \frac{1}{b} + V_2 \cdot \frac{1}{2b} \cdot \frac{b}{P} \\ &= P \cdot \frac{b}{2P} \cdot \frac{1}{b} + P \cdot \frac{1}{2b} \cdot \frac{b}{P} = \frac{1}{2} + \frac{1}{2} = 1 \leq k \quad \checkmark \end{aligned}$$

Reduction 1: Backward Direction (Part 1)

Backward ($\Leftarrow$): WTA has solution $\Rightarrow$ SUBSET PRODUCT has solution

Suppose WTA has solution with $D \leq 1$. Let X', Y' be weapons assigned to T_1, T_2.

Claim: W_α and W_β cannot both target the same target

Proof: If both assigned to T_1 (symmetric for T_2):

$$D \geq P \cdot \frac{1}{2b} \cdot \frac{b}{2P} \cdot \frac{1}{\prod_{x \in X' \setminus \{\alpha,\beta\}} a_x} + \frac{P}{\prod_{y \in Y'} a_y} \geq \frac{1}{4} + 1 > 1 \quad \Rightarrow\Leftarrow$$

So $W_\alpha \to T_1$ and $W_\beta \to T_2$. Let $X = X' \setminus \{\alpha\}$, $Y = Y' \setminus \{\beta\}$.

From the Assignment

With $W_\alpha \to T_1$ and $W_\beta \to T_2$:

$$D = P \cdot \frac{b}{2P} \prod_{x \in X} \frac{1}{a_x} + P \cdot \frac{1}{2b} \prod_{y \in Y} \frac{1}{a_y} \leq 1$$

Reduction 1: Backward Direction (Part 2)

Algebraic Derivation

Let $z = \prod_{x \in X} a_x$. Then:

$$\frac{b}{2z} + \frac{z}{2b} \leq 1$$

Multiply by $2bz > 0$:

$$b^2 + z^2 \leq 2bz \quad \Rightarrow \quad (b - z)^2 \leq 0$$

Since $(b - z)^2 \geq 0$, we must have $(b - z)^2 = 0$, so:

$$b = z = \prod_{x \in X} a_x$$

Conclusion

X is a valid solution to SUBSET PRODUCT! ✓
Time Complexity: $O(m)$ – Polynomial reduction complete.

Reduction 2: WTA $\leq_p$ Multiple Knapsack

Multiple Knapsack Construction

Given WTA with m targets, n weapons, probabilities $\{p_{i,j}\}$:

- **Knapsacks:** m knapsacks (one per target) with $C_j = \infty$
- **Items:** n items (one per weapon type)
- **Profit:** $v_{ij} = -\log(1 - p_{i,j}) = -\log(q_{i,j})$
- **Weight:** $w_{ij} = 0$ (or 1 if weapon limits exist)

Objective Transformation (Key Insight)

For single target j, minimizing $\prod q_{i,j}^{x_{i,j}}$ equals minimizing:

$$\log\left(\prod_{i=1}^{n} q_{i,j}^{x_{i,j}}\right) = \sum_{i=1}^{n} x_{i,j} \log(q_{i,j})$$

Equivalent to **maximizing**:

$$-\sum_{i=1}^{n} x_{i,j} \log(q_{i,j}) = \sum_{i=1}^{n} x_{i,j} \cdot v_{ij}$$

This is **exactly the Multiple Knapsack objective!**

Reduction 2: Constraint Mapping & Conclusion

Constraint Mapping

- **WTA constraint:** $\sum_{j=1}^{m} x_{i,j} \leq W_i$ (weapon type i used at most W_i times)
- **Maps to:** $\sum_{j=1}^{m} y_{ij} \leq 1$ (each item in at most one knapsack) when $W_i = 1$
- For general W_i: Create W_i copies of item i

Equivalence

Assignment $x_{i,j}$ optimal for WTA $\Leftrightarrow$ $y_{ij} = x_{i,j}$ optimal for Multiple Knapsack
Logarithmic transformation preserves ordering of solutions!
Time: $O(nm)$ to compute all $v_{ij} = -\log(1 - p_{i,j})$

Final Conclusion

WTA is in NP (solution verifiable in polynomial time) + **WTA is NP-hard**
$\Rightarrow$ **The Weapon-Target Assignment Problem is NP-complete!**

Comprehensive Algorithm Categories for WTA

Five major categories of solution approaches

Exact

- Optimal solutions
- Exponential time
- Small instances

Approximation

- Provable bounds
- Polynomial time
- Large instances

Randomized

- Probabilistic
- Quality $\propto$ samples
- Baseline approach

Heuristic

Fast, problem-specific, near-optimal

Metaheuristic

General-purpose, exploration/exploitation balance

Category 1: Exact Exponential Algorithms

Guarantee Optimal Solutions with Proven Bounds

These algorithms guarantee finding the optimal solution but have exponential time complexity.

Algorithm	Description	Complexity	Solution Quality
Branch-and-Bound (B&B)	Uses LP/IP relaxations with branching on weapon-target assignments. Employs maximum marginal return for branching strategy and network flow for lower bounds.	Worst-case: $O(2^{mn})$ nodes	Optimal; handles up to 80×80 instances
Branch-Price-and-Cut (BPC)	Reformulates WTA for column generation with efficient pricing problem solver. Uses clique cuts and power cone optimization for tight relaxations.	Polynomial per iteration; $O(nm \cdot K)$	Optimal; solves 10000+ in seconds
Branch-and-Adjust	Specialized B&B using compact piecewise linear convex under-approximation of objective. Built on MILP solvers with custom callbacks.	Exponential worst; poly approx	Optimal; 2% gap for 1500×1000 in 2h

Category 2: Approximation Algorithms

Provable Bounds with Polynomial Time Complexity

These algorithms provide theoretical guarantees on solution quality while running in polynomial time.

Algorithm	Description	Complexity	Solution Quality
Piecewise Linear Approximation	Approximates nonlinear survival probability using piecewise linear convex functions with breakpoints. Based on logarithmic transformation $-\log(1 - p_{ij})$.	$O(mn \cdot B)$ where $B =$ breakpoints	Provable δ-error bounds; suboptimal but efficient
LP Relaxation + Rounding	Relaxes integer constraints to LP, solves continuous problem, then rounds to feasible integer solution using various rounding schemes.	$O(\text{poly}(m, n))$ (interior point)	No approximation guarantee; may violate constraints
Lagrangian Relaxation	Relaxes weapon constraints via Lagrangian multipliers, decomposes into independent target subproblems. Uses subgradient method for multiplier updates.	$O(T \cdot nm)$ where $T =$ iterations	Provides lower bounds; typically 5-15% gap

Category 3: Randomized Algorithms

Probabilistic Methods for Solution Space Exploration

These algorithms use randomness to explore the solution space.

Algorithm	Description	Complexity	Solution Quality
Monte Carlo Sampling	Generates multiple random weapon-target assignments, evaluates each using survival probability function, selects best assignment.	$O(S \cdot mn)$ where $S =$ samples	Baseline; expected value $\rightarrow$ optimal as $S \rightarrow \infty$
Randomized Constructive Heuristic	Builds solutions by probabilistically selecting weapon-target pairs based on marginal returns with added randomness to escape local optima.	$O(R \cdot mn)$ where $R =$ restarts	Variable ($\pm$10-20%); depends on random seed

Problem-Specific Knowledge for Fast Good Solutions

Heuristics use domain knowledge to quickly find good (but not necessarily optimal) solutions.

Algorithm	Description	Complexity	Solution Quality
Maximum Marginal Return (MMR)	Greedy algorithm that sequentially assigns each weapon to target yielding maximum reduction in survival probability. Based on $\Delta_{ij} = V_j(q_j - q_j \cdot q_{ij})$.	$O(mn)$	Near-optimal; optimal when p_{ij} uniform; typically <5% from optimal
Very Large-Scale Neighborhood (VLSN)	Explores exponentially large neighborhoods via improvement graphs. Uses multi-exchange moves to escape local optima.	$O(m^2n^2)$ per iteration	High quality; improves MMR by 2-5%; effective for 200×200
MRBCH	Iteratively selects weapon-target-sensor triplets based on dynamic marginal return calculation. Updates threat values after each assignment.	$O(mn \log(mn))$	Fast; within 3-8% of optimal for sensor-WTA

General-Purpose Optimization Frameworks

Metaheuristics are adaptable frameworks that balance exploration and exploitation.

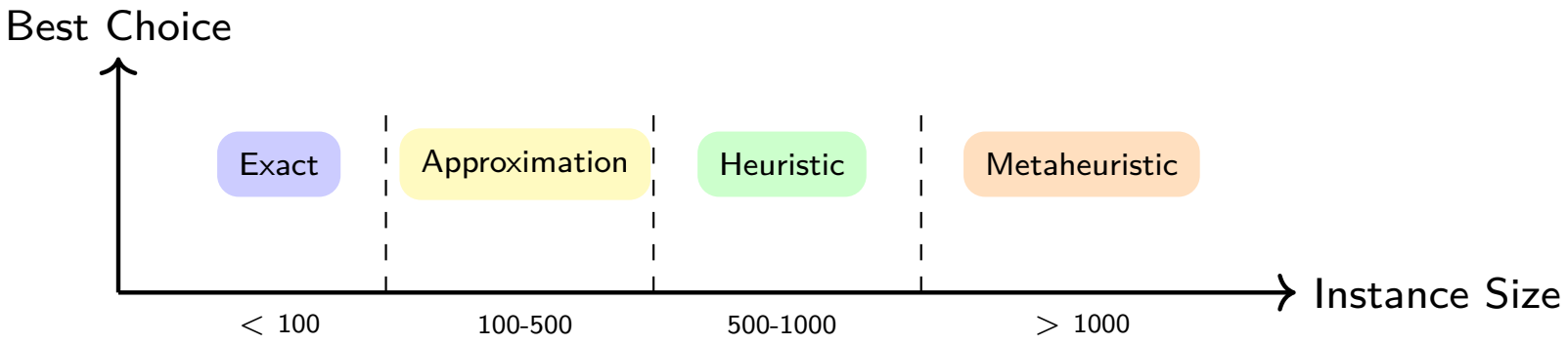
Algorithm	Description	Complexity	Solution Quality
Genetic Algorithm (GA)	Population-based evolution with crossover (weapon reassignment between parents) and mutation operators. Uses tournament selection and elitism.	$O(G \cdot P \cdot mn)$ G=gens, P=pop	Good for medium-large; 5-10% gap; may converge prematurely
Ant Colony Optimization (ACO)	Pheromone-based probabilistic construction. Transition probability: $P_{ij} = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum [\tau]^\alpha [\eta]^\beta}$	$O(I \cdot A \cdot mn)$ I=iters, A=ants	Competitive; avoids premature convergence; elite ant effective
Modified Pareto ACO (MPACO)	Multi-objective ACO for bi-objective WTA. Uses dynamic heuristic info, improved movement probability, dynamic evaporation rate, boundary mutation.	$O(I \cdot A \cdot mn \cdot K)$ K=objectives	Superior Pareto front; outperforms NSGA-II and SPEA-II

Complexity Comparison Summary

When to Use Each Category?

The choice depends on instance size, time constraints, and solution quality requirements.

Category	Typical Complexity	Instance Size Limit	Use Case
Exact	$O(2^{mn})$ - Exponential	< 1000 variables	When optimality required
Approximation	$O(\text{poly}(m, n))$	> 10000	Need theoretical bounds
Randomized	$O(S \cdot mn)$	Any (quality $\propto S$)	Baselines, diversity
Heuristic	$O(mn)$ - $O(m^2 n^2)$	> 1000	Real-time systems
Metaheuristic	$O(I \cdot P \cdot mn)$	> 10000	Large-scale optimization



Maximum Marginal Return (MMR) and Ant Colony Optimization (ACO)

Detailed algorithms with step-by-step examples

MMR – Greedy Heuristic

- Fast: $O(mn)$ complexity
- Deterministic output
- Near-optimal ($<$5% gap)
- **Ideal for real-time systems**

ACO – Metaheuristic

- Flexible: $O(I \cdot A \cdot mn)$
- Stochastic exploration
- Avoids local optima
- **Ideal for large instances**

Why MMR + ACO?

Maximum Marginal Return (MMR)

Why selected:

- $O(mn)$ – fastest complexity
- Simple to implement and verify
- Often near-optimal (<5% gap)
- Optimal when p_{ij} are uniform
- **Ideal for real-time missile defense**

Key Formula:

$$\Delta_{ij} = V_j \cdot q_j \cdot (1 - q_{ij})$$

(Marginal reduction in survival)

Ant Colony Optimization (ACO)

Why selected:

- Avoids premature convergence
- Natural exploration/exploitation balance
- Pheromone-based learning
- **Effective for large instances (>500)**
- Elite ant strategy improves quality

Key Formula:

$$P_{ij} = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum [\tau]^\alpha [\eta]^\beta}$$

(Transition probability)

Complementary Approaches

MMR provides fast initial solution; **ACO** refines for large-scale instances with multiple ants exploring solution space.

Algorithm 1: Maximum Marginal Return (MMR)

Core Idea

Greedy approach: At each step, assign the weapon-target pair that provides the **maximum marginal improvement** in expected damage.

Key Concepts

- **Combined Kill Probability:** When multiple weapons assigned to same target:

$$P_{kill} = 1 - \prod_{i \in S} (1 - p_i)$$

  where S is the set of weapons assigned to that target
- **Expected Value:** $\text{EV} = V_j \times P_{kill,j}$
- **Marginal Return:** $MR_{ij} = V_j \times P'_{kill,j} - V_j \times P_{kill,j}$

Example: Diminishing Returns

If W1 ($p = 0.7$) and W3 ($p = 0.8$) both assigned to T1:
$P_{kill} = 1 - (1 - 0.7) \times (1 - 0.8) = 1 - 0.06 = 0.94$

MMR Algorithm: Pseudocode

Algorithm 1 Maximum Marginal Return (MMR) Heuristic for WTA

```
Input: Weapons W, Targets T, Kill Prob P, Threat Values V
Output: Assignment A
Initialize A ← ∅ (no assignments)
Initialize P_kill[j] ← 0 for all targets j
while unassigned weapons remain do
    maxMarginalReturn ← 0
    bestPair ← null
    for each unassigned weapon i do
        for each target j do
            Calculate P'_{kill,j} (kill prob with weapon i added)
            MR_ij = V_j × P'_{kill,j} − V_j × P_{kill,j}
            if MR_ij > maxMarginalReturn then
                maxMarginalReturn ← MR_ij, bestPair ← (i, j)
            end if
        end for
    end for
    Assign weapon i to target j from bestPair
    Update P_{kill,j} and mark weapon i as assigned
end while
return A
```

MMR Example: Problem Setup (Naval Air Defense)

Incoming Missiles (Targets):

- T1: Aircraft Carrier ($V = 10$)
- T2: Destroyer ($V = 6$)
- T3: Supply Ship ($V = 3$)

Defensive Weapons:

- W1, W2, W3, W4 (4 interceptors)

Goal: Maximize expected damage prevented

Kill Probability Matrix P_{ij}:

	T1	T2	T3
W1	0.7	0.8	0.9
W2	0.6	0.7	0.8
W3	0.8	0.6	0.7
W4	0.5	0.9	0.6

Objective Function:

$$\text{Max} \sum_j V_j \times P_{kill,j}$$

MMR Iteration 1: Calculate Initial Marginal Returns

Algorithm Step: Calculate marginal return for all weapon-target pairs

Formula: $MR_{ij} = V_j \times P_{ij}$ (since no weapons assigned yet)

	T1 ($V = 10$)	T2 ($V = 6$)	T3 ($V = 3$)
W1	$10 \times 0.7 = 7.0$	$6 \times 0.8 = 4.8$	$3 \times 0.9 = 2.7$
W2	$10 \times 0.6 = 6.0$	$6 \times 0.7 = 4.2$	$3 \times 0.8 = 2.4$
W3	$10 \times 0.8 = \mathbf{8.0}$	$6 \times 0.6 = 3.6$	$3 \times 0.7 = 2.1$
W4	$10 \times 0.5 = 5.0$	$6 \times 0.9 = 5.4$	$3 \times 0.6 = 1.8$

MMR Choice

Highest Marginal Return = 8.0 $\Rightarrow$ Assign **W3** $\rightarrow$ **T1**

Current State: T1: $P_{kill} = 0.8$, Expected $= 8.0$ **Remaining:** W1, W2, W4

MMR Iteration 2: Recalculate with Diminishing Returns

Algorithm Step: Update marginal returns accounting for W3 already on T1

For T1 (already has W3, $P_{kill} = 0.8$):

- W1: $P'_{kill} = 1 - (0.3 \times 0.2) = 0.94$, $MR = 10(0.94 - 0.8) = 1.4$
- W2: $P'_{kill} = 1 - (0.4 \times 0.2) = 0.92$, $MR = 10(0.92 - 0.8) = 1.2$
- W4: $P'_{kill} = 1 - (0.5 \times 0.2) = 0.90$, $MR = 10(0.90 - 0.8) = 1.0$

	T1 (has W3)	T2 (empty)	T3 (empty)
W1	1.4	4.8	2.7
W2	1.2	4.2	2.4
W4	1.0	**5.4**	1.8

MMR Choice

Highest Marginal Return = 5.4 $\Rightarrow$ Assign **W4** $\rightarrow$ **T2**

Current State: T1: 8.0, T2: $6 \times 0.9 = 5.4$ **Total: 13.4**

MMR Iterations 3 & 4: Complete Assignment

Iteration 3: Marginal returns for W1, W2

	T1 (has W3)	T2 (has W4)	T3 (empty)
W1	1.4	0.48	**2.7**
W2	1.2	0.42	2.4

$\Rightarrow$ Assign **W1** $\rightarrow$ **T3** (MR $= 2.7$)

Iteration 4: Only W2 remains

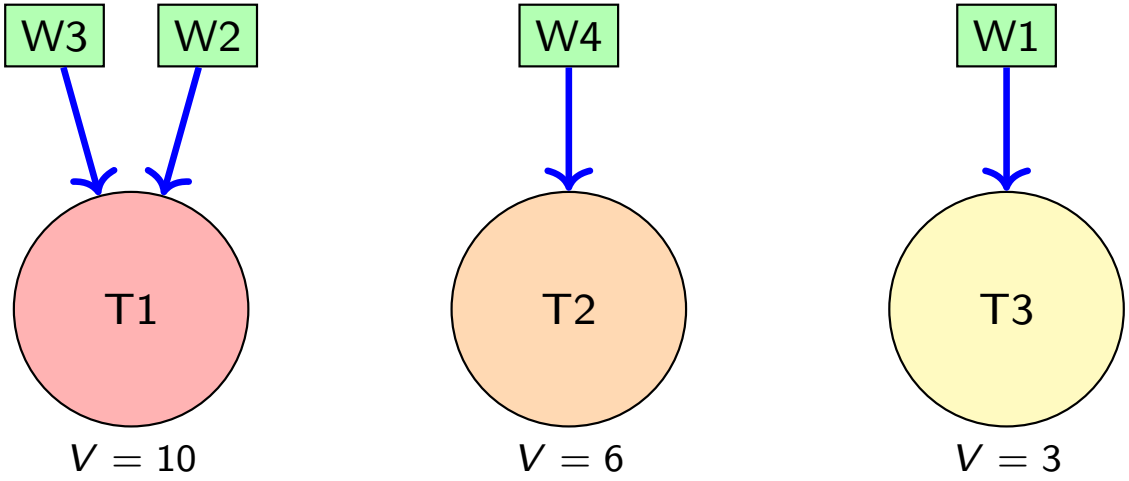
	T1 (has W3)	T2 (has W4)	T3 (has W1)
W2	**1.2**	0.42	0.24

$\Rightarrow$ Assign **W2** $\rightarrow$ **T1** (MR $= 1.2$)

Note: Diminishing Returns

T3 with W1 has $P_{kill} = 0.9$, so adding W2: $P'_{kill} = 1 - (0.2 \times 0.1) = 0.98$, $MR = 3(0.98 - 0.9) = 0.24$

MMR: Final Solution



Final Assignment & Expected Value

- **T1 (Aircraft Carrier):** W2, W3 $\rightarrow P_{kill} = 1 - (0.4 \times 0.2) = 0.92$, EV $= 10 \times 0.92 = 9.2$
- **T2 (Destroyer):** W4 $\rightarrow P_{kill} = 0.9$, EV $= 6 \times 0.9 = 5.4$
- **T3 (Supply Ship):** W1 $\rightarrow P_{kill} = 0.9$, EV $= 3 \times 0.9 = 2.7$

Total Expected Value (Damage Prevented)

$9.2 + 5.4 + 2.7 = \mathbf{17.3}$

MMR: Algorithm Analysis

Advantages

- **Fast:** $O(M^2 \cdot N)$ time complexity
- **Simple:** Easy to implement
- **Real-time capable:** Suitable for time-critical defense systems
- **Good results:** Often near-optimal
- **Adaptive:** Considers current state when making decisions

Disadvantages

- **No optimality guarantee:** May miss globally optimal solution
- **Greedy nature:** Early decisions can't be reversed
- **Local optimization:** Focuses on immediate best choice

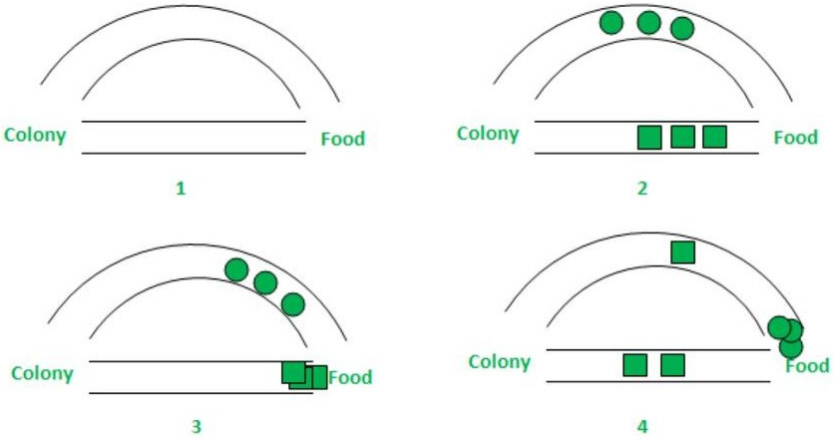
When to Use MMR

- Speed is critical
- Near-optimal is acceptable
- Real-time defense systems
- Large-scale problems

Algorithm 2: Ant Colony Optimization (ACO)

Core Idea

Bio-inspired approach: Ants deposit pheromones on good paths, guiding future ants toward better solutions.



Ants find shortest path via pheromone learning

Key Formulas

- **Selection:**

$$P_{ij} = \frac{[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta}{\sum_l [\tau_{il}]^\alpha \cdot [\eta_{il}]^\beta}$$

- **Evaporation:** $\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij}$
- **Deposit:** $\tau_{ij} \leftarrow \tau_{ij} + Q/F_k$

Pheromone Influence

If (W_1, T_1) has $\tau = 0.5$ and (W_1, T_2) has $\tau = 0.1$: Ant is $5\times$ more likely to choose T_1

ACO Algorithm: Pseudocode

Algorithm 2 Ant Colony Optimization (ACO) Metaheuristic for WTA

```
Initialize pheromone matrix τ_ij ← τ_0 for all (i, j)
Set parameters: α, β, ρ, Q, n_ants, n_iter
for iteration = 1 to n_iter do
    for ant k = 1 to n_ants do
        Initialize solution S_k ← ∅
        for weapon i = 1 to m do
            Calculate probability for each available target j:
                p_ij^k = [τ_ij]^α · [η_ij]^β / Σ_{l∈available} [τ_il]^α · [η_il]^β
                where η_ij = P_ij × V_j (heuristic information)
            Select target j using roulette wheel selection
            Add assignment (i, j) to S_k
        end for
        Calculate fitness F_k = Σ_{(i,j)∈S_k} P_ij × V_j
    end for
    Pheromone Update:
       Evaporation: τ_ij ← (1 − ρ) · τ_ij for all (i, j)
       Deposit: τ_ij ← τ_ij + Σ_{k:(i,j)∈S_k} Q / F_k
end for
return Best solution found
```

ACO Example: Problem Setup

3 Weapons, 3 Targets

Kill Probability Matrix (P_{ij}):

	T_1	T_2	T_3
W_1	0.7	0.5	0.4
W_2	0.6	0.8	0.5
W_3	0.5	0.6	0.9

Target Values:

- $V(T_1) = 100$ (high-value)
- $V(T_2) = 80$ (medium-value)
- $V(T_3) = 60$ (lower-value)

ACO Parameters:

Parameter	**Value**
Number of ants (n_{ants})	5
Number of iterations (n_{iter})	3
Pheromone importance (α)	1
Heuristic importance (β)	2
Evaporation rate (ρ)	0.5
Pheromone constant (Q)	1
Initial pheromone (τ_0)	0.1

ACO: Heuristic Matrix Initialization

Heuristic Information: $\eta_{ij} = P_{ij} \times V_j$

	T_1 $(V = 100)$	T_2 $(V = 80)$	T_3 $(V = 60)$
W_1	$0.7 \times 100 = 70$	$0.5 \times 80 = 40$	$0.4 \times 60 = 24$
W_2	$0.6 \times 100 = 60$	$0.8 \times 80 = 64$	$0.5 \times 60 = 30$
W_3	$0.5 \times 100 = 50$	$0.6 \times 80 = 48$	$0.9 \times 60 = 54$

Initial Pheromone Matrix: All entries $= \tau_0 = 0.1$

Interpretation

Higher η_{ij} values indicate more attractive weapon-target assignments based on **expected damage**.

ACO Iteration 1: Ant 1 Assigns W_1

Available targets: $\{T_1, T_2, T_3\}$

Calculate attractiveness: $[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta$ with $\alpha = 1$, $\beta = 2$

$$\begin{aligned}
\text{For } T_1: &\quad (0.1)^1 \times (70)^2 = 0.1 \times 4900 = 490 \\
\text{For } T_2: &\quad (0.1)^1 \times (40)^2 = 0.1 \times 1600 = 160 \\
\text{For } T_3: &\quad (0.1)^1 \times (24)^2 = 0.1 \times 576 = 57.6 \\
\text{Sum } \Sigma: &\quad 707.6
\end{aligned}$$

Selection Probabilities:

- $P(T_1) = 490/707.6 =$ **69.2%**
- $P(T_2) = 160/707.6 = 22.6\%$
- $P(T_3) = 57.6/707.6 = 8.1\%$

Selection (Roulette Wheel)

$W_1 \rightarrow T_1$ (highest probability selected)

ACO Iteration 1: Ant 1 Completes Solution

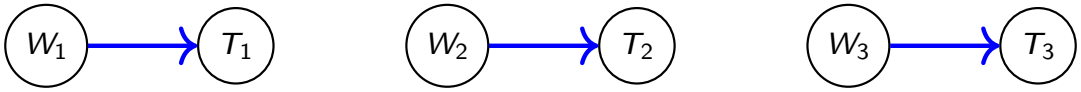
Assigning W_2: Available targets $\{T_2, T_3\}$ (T1 taken)

$$\text{For } T_2: \quad (0.1)^1 \times (64)^2 = 409.6 \quad \Rightarrow P = 82.0\%$$
$$\text{For } T_3: \quad (0.1)^1 \times (30)^2 = 90 \quad \Rightarrow P = 18.0\%$$

$\Rightarrow W_2 \rightarrow T_2$

Assigning W_3: Only T_3 remains $\Rightarrow W_3 \rightarrow T_3$

Ant 1 Solution & Fitness



$F_1 = 100 \times 0.7 + 80 \times 0.8 + 60 \times 0.9 = 70 + 64 + 54 = \mathbf{188}$

ACO Iteration 1: All Ants Complete

All 5 ants construct solutions using probabilistic selection:

Ant	Solution	Fitness
1	$W_1 \to T_1, W_2 \to T_2, W_3 \to T_3$	**188**
2	$W_1 \to T_2, W_2 \to T_1, W_3 \to T_3$	154
3	$W_1 \to T_1, W_2 \to T_3, W_3 \to T_2$	148
4	$W_1 \to T_1, W_2 \to T_2, W_3 \to T_3$	**188**
5	$W_1 \to T_3, W_2 \to T_2, W_3 \to T_1$	138

Best solutions: Ant 1 and Ant 4 (Fitness = 188)

Key Insight

Different ants explore different solutions due to **probabilistic selection**, allowing exploration of solution space!

ACO: Pheromone Update

Step 1: Evaporation ($\rho = 0.5$)

$$\tau_{ij}^{new} = (1 - \rho) \times \tau_{ij}^{old}$$

All pheromones: $0.1 \rightarrow 0.05$

	T_1	T_2	T_3
W_1	0.05	0.05	0.05
W_2	0.05	0.05	0.05
W_3	0.05	0.05	0.05

Step 2: Deposit ($\Delta\tau = Q/F_k$)

Each ant deposits on its path:

- Ant 1: $\Delta\tau = 1/188 = 0.0053$
- Ant 2: $\Delta\tau = 1/154 = 0.0065$
- Ant 3: $\Delta\tau = 1/148 = 0.0068$
- Ant 4: $\Delta\tau = 1/188 = 0.0053$
- Ant 5: $\Delta\tau = 1/138 = 0.0072$

Example: (W_1, T_1) Calculation

- After evaporation: 0.05
- Ant 1 deposits: $+0.0053$
- Ant 3 deposits: $+0.0068$
- Ant 4 deposits: $+0.0053$
- **Total:** $= \mathbf{0.0674}$

Updated Pheromone Matrix

	T_1	T_2	T_3
W_1	0.0674	0.0565	0.0572
W_2	0.0565	0.0691	0.0568
W_3	0.0572	0.0568	0.0693

Best paths have higher pheromones!

ACO: Convergence After Multiple Iterations

Process repeats:

1. Ants construct solutions using **updated pheromones**
2. Now more likely to choose high-pheromone paths
3. Good solutions reinforced, bad solutions fade

After 3 iterations:

- Pheromones converge on optimal assignment
- (W_1, T_1), (W_2, T_2), (W_3, T_3) have much higher pheromones
- Most ants select this solution

Final Best Solution

$$\boxed{W_1 \to T_1, \quad W_2 \to T_2, \quad W_3 \to T_3}$$

Fitness = 188 (Maximum possible for this instance!)

ACO: Algorithm Analysis

Advantages

- **Time complexity:** $O(n_{iter} \times n_{ants} \times n^2)$
- **Handles large-scale:** Effective for many weapons/targets
- **Near-optimal:** Finds high-quality solutions without exhaustive search
- **Parallelizable:** Multiple ants work independently
- **Adaptive:** Learns good assignments through pheromone trails
- **Robust:** Avoids local optima via probabilistic exploration

Disadvantages

- **Parameter tuning:** Requires careful selection of α, β, ρ, Q
- **Slower than greedy:** Multiple iterations needed
- **Stochastic:** Different runs may yield different results

When to Use ACO

- Large-scale instances (>500)
- Time permits iterations
- High solution quality needed
- Escaping local optima critical

How We Will Evaluate Our Implementations

Comprehensive experimental methodology for algorithm comparison

Algorithms

- MMR (Greedy)
- ACO (Meta)
- B&B (Exact)

Instances

- 5 size categories
- 30 instances each
- 150 total tests

Metrics

- Quality (gap)
- Runtime
- Scalability

Algorithm Comparison Overview

Three Algorithmic Approaches

We implement and compare three fundamentally different approaches to the WTA problem:

Algorithm	Type	Complexity	Characteristics
MMR	Greedy heuristic	$O(W^2 \cdot T)$	Fast, deterministic
ACO	Metaheuristic	$O(k \cdot A \cdot W \cdot T)$	Good quality, flexible
B&B	Exact	$O(T^W)$ worst-case	Optimal but slow

Expected Tradeoffs

- **MMR:** Real-time applications, potentially lower quality but fastest execution
- **ACO:** Balance between quality and time; suitable for medium-large instances
- **B&B:** Optimal solutions with proof; limited to small instances due to exponential growth

Benchmark Instances: Test Problem Sizes

Five Categories of Problem Complexity

We test across a range of instance sizes to evaluate scalability.

Category	**Weapons**	**Targets**	**Variables ($W \times T$)**	**Instances**
Small	5	5	25	30
Small-Medium	10	10	100	30
Medium	20	20	400	30
Medium-Large	50	50	2,500	30
Large	100	100	10,000	30

Total: 150 Benchmark Instances

Each category has 30 randomly generated instances for statistical significance.

Instance Generation Protocol

Random Parameter Generation

Target Values:

$$v_j \sim \text{Uniform}(50, 100)$$

Higher values = more important targets

Kill Probabilities:

$$p_{ij} \sim \text{Uniform}(0.3, 0.9)$$

Realistic range for weapon effectiveness

Weapon Ammunition:

$$\sum_i \mu_i = 2 \times \text{num_weapons}$$

Total ammo = twice the weapon count

Storage Format

CSV Format:

- `instance_id`: Unique identifier
- `num_weapons`: W value
- `num_targets`: T value
- `values`: Comma-separated v_j
- `probabilities`: $W \times T$ matrix
- `ammo`: Per-weapon limits

Reproducibility:

- Fixed random seeds
- Stored instance files

Algorithm Parameter Settings

Configuration for Fair Comparison

All parameters are fixed across all instances to ensure fair comparison.

Algorithm	Parameters	Rationale
MMR	Deterministic (no tuning needed)	Greedy nature requires no parameters
ACO	$\alpha = 1.0$ (pheromone weight)	Standard ACO settings from literature
	$\beta = 2.0$ (heuristic weight)	Emphasizes problem-specific info
	$\rho = 0.1$ (evaporation rate)	Slow evaporation for exploration
	max_iter $= 1000$	Sufficient for convergence
	$n_{ants} = 20$	Balance quality/speed
B&B	Time limit: 3600s (1 hour)	Practical timeout for experiments
	MIP gap: 0.01%	Near-optimal termination

Parameter Range Experimentation

We will test a range of parameter values (α, β, ρ, iterations, ants) to analyze sensitivity and optimize performance across different instance sizes.

Performance Metrics: Solution Quality

Objective Function Value

Minimize expected survival value:

$$f(x) = \sum_{j=1}^{n} v_j \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$$

Lower $f(x)$ = better solution (more damage inflicted)

Optimality Gap

Definition:

$$\text{Gap} = \frac{f_{\text{alg}} - f_{\text{opt}}}{f_{\text{opt}}} \times 100\%$$

Interpretation:

- Gap = 0% means optimal
- Gap = 5% means 5% worse than optimal
- B&B provides f_{opt} baseline

Quality Comparisons

Primary Questions:

1. How close is MMR to optimal?
2. Does ACO improve over MMR?
3. At what size does B&B timeout?

Expected Results:

- MMR: 3-8% gap
- ACO: 1-5% gap
- B&B: 0% gap (when terminates)

Performance Metrics: Computational Efficiency

Runtime Metrics

Primary:

- Wall-clock time (seconds)
- CPU time (for accuracy)

Secondary:

- ACO: Iterations to convergence
- B&B: Nodes explored
- B&B: Timeout count

Scalability Limits

Algorithm	Expected Limit
MMR	$\sim 1000 \times 1000$
ACO	$\sim 500 \times 500$
B&B	$\sim 50 \times 50$

Statistical Analysis

Summary Statistics:

- Mean, median, std deviation
- Min/max values
- Quartiles for box plots

Significance Testing:

- Wilcoxon signed-rank test
- Pairwise comparisons (MMR vs ACO, ACO vs B&B)
- $p < 0.05$ for significance

Over 30 instances per category:
Ensures statistical validity

Implementation Pipeline

Phase 1: Core

Implementation

1. Implement MMR greedy heuristic
2. Implement ACO metaheuristic
3. Implement B&B exact algorithm
4. Unit tests for correctness

Phase 2: Experiments

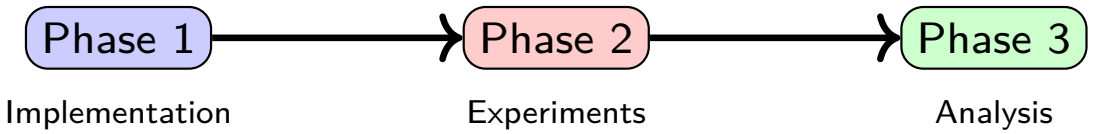
Experimentation

1. Generate 150 benchmark instances (CSV)
2. Execute all algorithms
3. Log: instance_id, algorithm, objective, runtime, gap
4. Handle timeouts

Phase 3: Analysis

Visualization

1. Compute summary statistics
2. Create 6 comparison plots
3. Statistical significance tests
4. Final report



Tools and Libraries

Programming Environment

Language: Python 3.11

Core Libraries:

- `numpy`: Numerical operations, matrix manipulation
- `pandas`: Data management, CSV I/O, result aggregation
- `gurobipy`: MILP solver for B&B exact method

Visualization:

- `matplotlib`: Publication-quality plots
- `seaborn`: Statistical visualizations

Analysis Libraries

Statistics:

- `scipy.stats`: Wilcoxon test, normality tests
- `statsmodels`: Advanced statistical analysis

Output Deliverables:

- Results CSV with all metrics
- 6 comparison plots (quality, time, scalability, gaps)
- Statistical significance table
- Algorithm source code

Visualization Plan

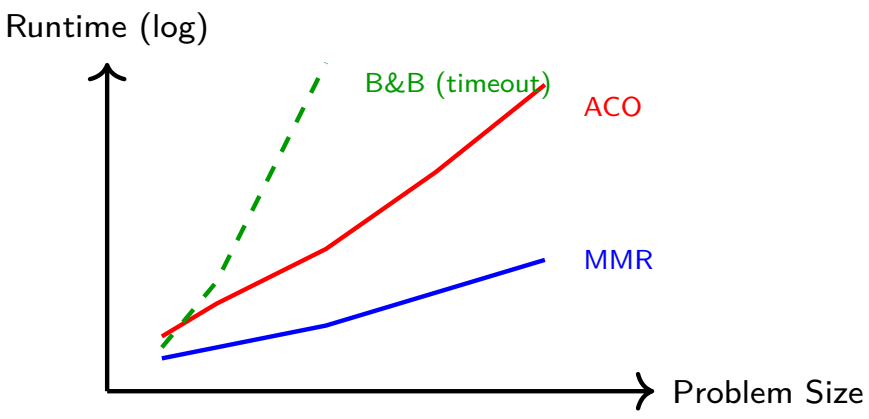
Six Plots to Generate for Final Analysis:

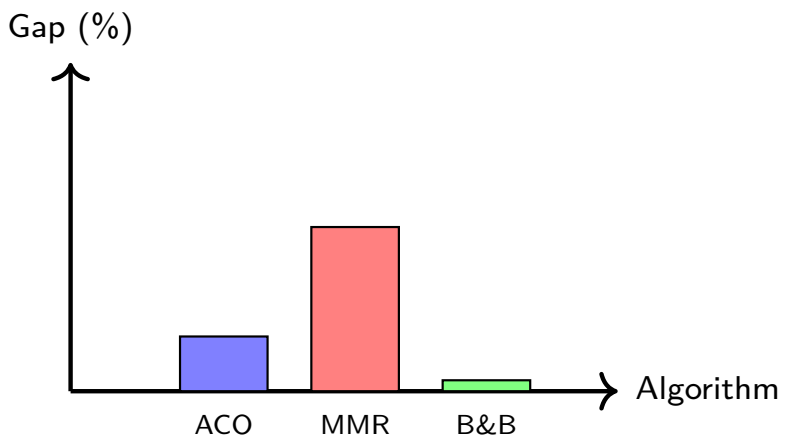
Quality Analysis

1. **Box Plot:** Objective value distribution by algorithm per problem size
2. **Line Plot:** Mean solution quality vs. problem size
3. **Gap Chart:** Optimality gap comparison across algorithms

Performance Analysis

4. **Bar Chart:** Average runtime comparison
5. **Log-Log Plot:** Scalability curves (runtime vs. size)
6. **Stats Table:** Wilcoxon test p-values (pairwise)





Real-World Applications of WTA

Military and Civilian Uses of Weapon-Target Assignment

WTA is not just a military problem – it's a fundamental optimization problem!

The WTA framework applies to any situation involving **limited resources**, **multiple objectives**, and **probabilistic outcomes**.

Why is WTA Important?

Theoretical Importance

- **NP-Complete** problem – fundamental complexity class
- Test case for algorithm research
- Used in **Game Theory** and strategic analysis
- Drives optimization innovations
- Connections to Knapsack, Assignment, Covering problems

Practical Importance

- Military + Civilian applications
- **Saves lives** in emergency response
- **Saves resources** through optimal allocation
- Real-time decision making under uncertainty
- Growing relevance with autonomous systems

Military Applications: Air Defense Systems

Example: US Navy AEGIS System & Israel's Iron Dome

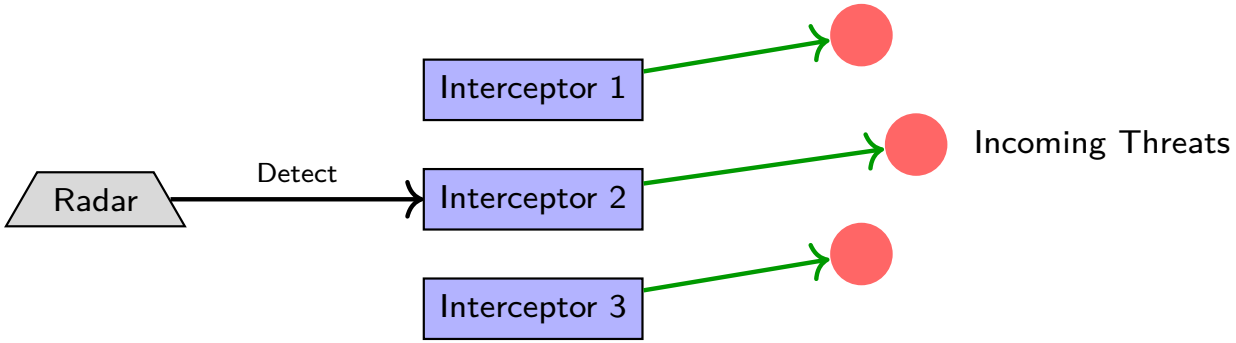
Incoming missiles must be intercepted with limited defensive weapons.

The Challenge

- Enemy missiles incoming – which interceptor targets which missile?
- **Time constraint:** Only **seconds** to decide
- **Resource constraint:** Limited interceptor missiles
- **Uncertainty:** Kill probability $< 100\%$

WTA Solution

- **Weapons:** Interceptor missiles
- **Targets:** Incoming threats
- **Objective:** Maximize expected kill probability
- **Result:** 90%+ interception rate (Iron Dome)



WTA assigns interceptors to maximize survival

Military Applications: Counter-Drone Systems

Emerging Threat: Drone Swarms

Modern warfare involves coordinated drone attacks that overwhelm traditional defenses.

Available Countermeasures

- **Electronic Jamming:** Disrupt communications
- **Kinetic Weapons:** Physical destruction
- **Directed Energy:** Lasers, microwave
- **Counter-Drones:** Autonomous interceptors

WTA for Layered Defense

- Different weapons have different kill probabilities against different drone types
- **Multi-objective:** Cost vs. effectiveness
- **Dynamic:** Real-time reallocation as situation evolves
- Hierarchical WTA for layered engagement

Non-Military Applications – Part 1

Emergency Response & Disaster Management

Scenario: During disasters – where to send ambulances, fire trucks?

WTA Mapping:

- **Weapons** = Ambulances, Fire Trucks, Rescue Teams
- **Targets** = Emergency Incidents, Fire Locations, Trapped Victims
- **Kill Probability** = Success rate based on distance, severity
- **Goal** = Minimize casualties, maximize lives saved

Healthcare – Hospital ER Triage

Scenario: Hospital ER – which doctor treats which patient?

WTA Mapping:

- **Weapons** = Doctors, Nurses, Equipment, Operating Rooms
- **Targets** = Patients with varying conditions
- **Kill Probability** = Treatment success rate
- **Goal** = Maximize expected lives saved (QALY)

Non-Military Applications – Part 2

Cybersecurity – Network Defense

Scenario: Network under attack – which defense against which threat?

WTA Mapping:

- **Weapons** = Firewalls, IDS/IPS, Honeypots, Analysts
- **Targets** = Intrusions, Malware, DDoS attacks
- **Kill Probability** = Detection/blocking rate
- **Goal** = Block maximum attacks, protect high-value assets

Advertising & Marketing

Scenario: Which ad to show to which audience?

WTA Mapping:

- **Weapons** = Ad Placements (TV, Facebook, Google)
- **Targets** = Audience Segments (demographics)
- **Kill Probability** = Conversion rate per segment
- **Goal** = Maximum reach/conversion with budget

More Real-World Applications

Conservation

- Wildlife protection
- Anti-poaching patrol allocation
- Nature reserve selection
- Species preservation resources

Search & Rescue (SAR)

- Lost person search operations
- Maritime SAR
- Team allocation to sectors
- Probabilistic area coverage

Transportation & Logistics

- Vehicle routing
- Delivery optimization
- Fleet management
- Airport gate assignment

Common Pattern Across All Applications

All these problems involve: **Limited Resources** + **Multiple Objectives** + **Probabilistic Outcomes** + **Value Prioritization**

Computational Challenges in Real Applications

Time Constraints Vary by Domain

- **Missile Defense:** Only a few **seconds**!
- **Emergency Response:** A few **minutes**
- **Healthcare Triage:** **Hours**
- **Advertising Planning:** **Days to weeks**

⇒ **Different time constraints require different algorithms!**

Computational Problems

1. **NP-Complete:** Finding optimal solution takes exponential time
2. **Large Scale:** Thousands of variables possible (100+ weapons × 100+ targets)
3. **Uncertainty:** Must handle probabilistic outcomes
4. **Real-time:** Some applications need instant decisions
5. **Dynamic:** Situation changes during execution

Solution: Algorithm Selection Based on Time Budget

Real-time (seconds): Use **MMR** greedy heuristic

Planning (minutes-hours): Use **ACO** or **B&B**

Key Takeaways: WTA Applications

Universal Problem:

- WTA is not just for warfare – used in emergency services, healthcare, advertising, conservation, cybersecurity, transportation

Computational Challenge:

- NP-Complete problem – exact solution very difficult for large scale
- Different time constraints require different algorithmic approaches

Real-World Impact:

- **Saves lives** in missile defense and emergency response
- **Saves money** in resource allocation and advertising
- **Saves time** through automated decision making

Algorithm Research:

- Drives development of new optimization techniques (metaheuristics, approximation algorithms)
- Connections to other NP-hard problems enable knowledge transfer

Summary

1. **Problem:** WTA is a fundamental NP-complete optimization problem
2. **Hardness:** Proven via reductions from SUBSET PRODUCT
3. **Algorithms:** Surveyed exact, approximation, heuristic, metaheuristic
4. **Implementation:** MMR (greedy) + ACO (metaheuristic) + B&B (exact)
5. **Experiments:** 150 instances, multiple metrics, statistical analysis
6. **Applications:** Military defense, healthcare, emergency, cybersecurity

Next Steps

- Complete Python implementation
- Run all benchmark experiments
- Generate visualizations & report

Expected Outcomes

- MMR: Fast, $<5\%$ gap
- ACO: Better quality, slower
- B&B: Optimal for small instances

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Thank You!

Questions?